

COOK, MN, USA

WMO: 726549

Lat: 47.822N

Lon: 92.689W

Elev: 1327

StdP: 14.00

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54905

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-18.5	-32.7	0.9	-26.3	-25.9	1.4	-19.4	18.1	12.6	16.5	12.9	1.1	270	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	84.3	69.3	81.8	67.5	79.4	65.7	72.9	80.9	70.5	78.7	68.4	76.1	8.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.3	117.2	78.1	67.6	106.7	75.5	65.7	99.6	73.5	37.3	81.0	35.1	79.0	33.5	76.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.9	12.5	-32.0	89.7	5.0	2.6	-35.6	91.5	-38.5	93.1	-41.3	94.5	-44.9	96.4		
(5)			-32.0	75.8	4.9	1.7	-35.5	77.0	-38.4	78.0	-41.2	79.0	-44.8	80.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.8	8.2	11.9	25.0	38.9	51.5	60.3	65.7	63.2	55.5	42.3	28.1	13.7		
	DBStd	22.66	14.50	14.23	13.54	9.65	8.75	6.79	5.94	6.14	8.69	8.79	11.22	13.30		
	HDD50	5684	1296	1068	778	351	84	5	0	1	42	274	660	1125		
	HDD65	9751	1761	1488	1240	784	428	172	63	109	303	705	1108	1590		
	CDD50	1607	0	0	3	17	132	313	488	409	208	35	2	0		
	CDD65	200	0	0	0	0	11	30	87	52	19	1	0	0		
	CDH74	2240	0	0	0	7	168	360	947	563	182	13	0	0		
	CDH80	483	0	0	0	1	33	67	229	117	34	2	0	0		
	WSAvg	4.8	4.8	4.7	5.0	5.2	5.3	4.5	4.1	4.0	4.8	5.2	5.1	4.4		
	PrecAvg	27.4	0.7	0.7	1.1	2.1	3.1	4.3	3.9	3.3	3.4	2.6	1.4	0.9		
	PrecMax	36.0	1.8	1.9	3.1	5.4	6.5	7.6	11.1	5.4	8.1	5.2	3.6	1.6		
(15) Precipitation	PrecMin	20.2	0.2	0.1	0.4	0.6	1.1	1.1	1.6	1.2	0.7	0.8	0.1	0.2		
	PrecStd	3.5	0.4	0.4	0.7	1.1	1.3	1.5	2.0	1.3	1.7	1.1	0.9	0.4		
	DB	40.7	46.4	63.7	73.4	83.9	86.1	89.9	88.1	84.0	74.9	61.0	42.9			
	MCWB	35.8	39.5	52.5	55.6	65.0	70.0	72.8	72.6	67.8	59.5	51.7	37.5			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	35.9	42.6	54.9	66.5	79.2	81.4	84.4	82.3	79.3	68.0	54.3	37.2			
	2% MCWB	33.3	37.0	45.7	50.7	61.4	66.4	69.9	68.9	66.1	56.5	48.9	34.6			
	5% DB	32.0	36.8	48.3	63.1	73.2	78.8	81.9	80.7	74.8	62.9	47.9	34.4			
	5% MCWB	30.8	33.7	41.9	48.2	57.5	64.2	68.3	67.7	64.4	53.9	42.9	32.7			
	10% DB	28.0	33.8	43.4	57.1	69.8	74.7	79.4	76.9	71.9	56.9	44.6	31.8			
	10% MCWB	26.7	31.4	37.5	45.2	56.2	62.4	66.5	65.1	62.7	49.7	40.6	30.9			
	0.4% WB	36.7	41.8	57.0	58.0	69.0	73.3	75.9	75.0	71.9	63.0	54.4	38.1			
	0.4% MCDB	40.0	45.7	61.1	70.3	78.3	82.0	83.9	81.8	80.1	68.9	57.2	40.6			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	33.6	37.4	47.6	53.2	64.5	69.7	73.6	71.9	68.8	58.2	48.0	35.5			
	2% MCDB	35.2	41.2	52.9	62.2	73.1	77.3	81.0	80.5	75.3	66.5	54.0	37.6			
	5% WB	30.3	34.2	42.5	49.9	61.0	67.1	70.9	69.3	66.3	54.7	44.0	33.0			
	5% MCDB	31.6	37.0	48.6	61.0	70.3	74.7	79.0	77.3	72.5	61.2	47.3	34.5			
	10% WB	27.2	30.6	38.2	46.3	57.7	64.4	68.6	67.3	63.8	51.4	40.0	30.3			
	10% MCDB	27.9	33.2	42.9	55.9	65.6	71.5	77.0	74.3	69.3	55.9	43.9	31.4			
	MDBR	20.5	22.9	22.4	22.7	24.0	22.6	23.1	23.0	22.0	17.8	15.2	16.8			
	5% DB	18.6	21.6	24.9	32.2	31.8	26.7	25.9	26.4	26.6	26.7	22.1	17.6			
(36) Mean Daily Temperature Range	MCWBR	16.0	16.4	16.6	18.7	16.6	14.1	13.1	13.9	15.2	16.5	15.6	15.0			
	MCDBR	17.3	18.2	22.8	28.4	24.9	21.9	22.9	22.2	21.6	23.3	19.9	16.2			
	MCWBR	15.1	14.1	16.2	17.9	14.4	13.4	13.5	13.0	13.6	15.4	15.1	14.2			
	MCDBR	15.1	14.1	16.2	17.9	14.4	13.4	13.5	13.0	13.6	15.4	15.1	14.2			
(40) Clear-Sky Solar Irradiance	taub	0.251	0.264	0.314	0.345	0.359	0.372	0.390	0.373	0.356	0.307	0.282	0.260			
	taud	2.359	2.326	2.340	2.382	2.402	2.383	2.355	2.431	2.454	2.577	2.523	2.395			
	Ebn at Noon	268	288	287	288	287	282	275	276	270	271	256	249			
	Edh at Noon	22	28	33	35	36	37	37	33	30	22	19	19			
(44) All-Sky Solar Radiation	RadAvg	437	804	1180	1486	1665	1772	1895	1605	1189	702	399	309			
	RadStd	53	78	110	145	117	137	93	118	94	96	51	32			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page